



INTERNATIONAL ISLAMIC UNIVERSITY

SOFTWARE ENGINEERING DEPARTMENT

Total Marks: _____

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Project

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Project: Relational Database Implementation in SQL

Objective:

The objective of this final project is to transform your conceptual and logical database designs into a fully functional relational database using a Relational Database Management System (RDBMS) like Oracle. You will write the necessary SQL scripts to create, populate, and query your database based on the normalized schema developed in Assignment 03.

Part 1: Data Definition Language (DDL) Script

This script includes:

- DROP TABLE statements
- CREATE TABLE statements
- Primary Keys
- Foreign Keys
- UNIQUE constraints
- CHECK constraints
- NOT NULL constraints

-- DROP TABLES (Reverse Dependency Order)

DROP TABLE PAYMENT CASCADE CONSTRAINTS;
DROP TABLE ORDER_DETAILS CASCADE CONSTRAINTS;
DROP TABLE ORDERS CASCADE CONSTRAINTS;

DROP TABLE MENU_ITEM CASCADE CONSTRAINTS;
DROP TABLE CUSTOMER CASCADE CONSTRAINTS;

DROP TABLE WAITER CASCADE CONSTRAINTS;
DROP TABLE CHEF CASCADE CONSTRAINTS;

DROP TABLE SUPPLIER CASCADE CONSTRAINTS;
DROP TABLE EMPLOYEE CASCADE CONSTRAINTS;

DROP TABLE BRANCH CASCADE CONSTRAINTS;

```
CREATE TABLE BRANCH (  
    Branch_ID NUMBER PRIMARY KEY,  
    Branch_Name VARCHAR2(50) NOT NULL,  
    Address VARCHAR2(100) NOT NULL,  
    Phone VARCHAR2(15) UNIQUE  
);  
CREATE TABLE EMPLOYEE (  
    Employee_ID NUMBER PRIMARY KEY,  
    Branch_ID NUMBER NOT NULL,  
    Employee_Name VARCHAR2(50) NOT NULL,  
    Position VARCHAR2(30),  
    Salary NUMBER(10,2)  
        CHECK (Salary > 0),  
    Hire_Date DATE,  
  
    CONSTRAINT FK_EMP_BRANCH  
    FOREIGN KEY (Branch_ID)  
    REFERENCES BRANCH(Branch_ID)  
);
```



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-- CHEF

```
-----  
  
CREATE TABLE CHEF (  
    Chef_ID NUMBER PRIMARY KEY,  
    Employee_ID NUMBER UNIQUE,  
    Specialty VARCHAR2(50),  
  
    CONSTRAINT FK_CHEF_EMP  
    FOREIGN KEY (Employee_ID)  
    REFERENCES EMPLOYEE(Employee_ID)  
);
```

-- WAITER

```
-----  
  
CREATE TABLE WAITER (  
    Waiter_ID NUMBER PRIMARY KEY,  
    Employee_ID NUMBER UNIQUE,  
    Shift VARCHAR2(20)  
    CHECK (Shift IN ('Morning','Evening','Night')),  
  
    CONSTRAINT FK_WAITER_EMP  
    FOREIGN KEY (Employee_ID)  
    REFERENCES EMPLOYEE(Employee_ID)  
);
```

-- SUPPLIER

```
-----  
  
CREATE TABLE SUPPLIER (  
    Supplier_ID NUMBER PRIMARY KEY,  
    Branch_ID NUMBER NOT NULL,  
    Supplier_Name VARCHAR2(50) NOT NULL,  
    Phone VARCHAR2(15),  
  
    CONSTRAINT FK_SUPPLIER_BRANCH  
    FOREIGN KEY (Branch_ID)  
    REFERENCES BRANCH(Branch_ID)  
);
```

-- CUSTOMER

```
-----  
  
CREATE TABLE CUSTOMER (  
    Customer_ID NUMBER PRIMARY KEY,  
    Customer_Name VARCHAR2(50) NOT NULL,  
    Phone VARCHAR2(15) UNIQUE  
);
```

-- MENU ITEM



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```
-----  
CREATE TABLE MENU_ITEM (  
    Item_ID NUMBER PRIMARY KEY,  
    Item_Name VARCHAR2(50) NOT NULL,  
    Category VARCHAR2(30),  
    Price NUMBER(8,2)  
        CHECK (Price > 0)  
);
```

```
-----  
-- WAITER ORDERS  
-----
```

```
CREATE TABLE ORDERS (  
    Order_ID NUMBER PRIMARY KEY,  
    Customer_ID NUMBER NOT NULL,  
    Waiter_ID NUMBER NOT NULL,  
    Order_Date DATE,  
    Total_Amount NUMBER(10,2)  
        CHECK (Total_Amount >= 0),  
  
    CONSTRAINT FK_ORDER_CUSTOMER  
        FOREIGN KEY (Customer_ID)  
        REFERENCES CUSTOMER(Customer_ID),  
  
    CONSTRAINT FK_ORDER_WAITER  
        FOREIGN KEY (Waiter_ID)  
        REFERENCES WAITER(Waiter_ID)  
);
```

```
-----  
-- ORDER DETAILS  
-----
```

```
CREATE TABLE ORDER_DETAILS (  
    Order_Detail_ID NUMBER PRIMARY KEY,  
    Order_ID NUMBER NOT NULL,  
    Item_ID NUMBER NOT NULL,  
    Quantity NUMBER  
        CHECK (Quantity > 0),  
  
    CONSTRAINT FK_DETAIL_ORDER  
        FOREIGN KEY (Order_ID)  
        REFERENCES ORDERS(Order_ID),  
  
    CONSTRAINT FK_DETAIL_ITEM  
        FOREIGN KEY (Item_ID)  
        REFERENCES MENU_ITEM(Item_ID)  
);
```

```
-----  
-- PAYMENT  
-----
```



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```
CREATE TABLE PAYMENT (  
  Payment_ID NUMBER PRIMARY KEY,  
  Order_ID NUMBER UNIQUE,  
  Payment_Date DATE,  
  Amount NUMBER(10,2)  
    CHECK (Amount > 0),  
  Payment_Method VARCHAR2(20)  
    CHECK (Payment_Method IN ('Cash','Card','Online')),  
  
  CONSTRAINT FK_PAYMENT_ORDER  
  FOREIGN KEY (Order_ID)  
  REFERENCES ORDERS(Order_ID)  
);
```

Part 2: Data Manipulation Language (DML) Script

We must insert data in this order:

1. BRANCH
2. EMPLOYEE
3. CHEF
4. WAITER
5. SUPPLIER
6. CUSTOMER
7. MENU_ITEM
8. ORDERS
9. ORDER_DETAILS
10. PAYMENT

1. BRANCH

Query :

```
INSERT INTO BRANCH VALUES (1,'Blue Moon','F-8 Islamabad','0511111111');  
INSERT INTO BRANCH VALUES (2,'Royal Taste','G-9 Islamabad','0512222222');  
INSERT INTO BRANCH VALUES (3,'Food Street','I-8 Islamabad','0513333333');  
INSERT INTO BRANCH VALUES (4,'Spice Hub','F-10 Islamabad','0514444444');  
INSERT INTO BRANCH VALUES (5,'Tandoori House','G-10 Islamabad','0515555555');  
INSERT INTO BRANCH VALUES (6,'Biryani Point','I-9 Islamabad','0516666666');  
INSERT INTO BRANCH VALUES (7,'Urban Eatery','H-8 Islamabad','0517777777');  
INSERT INTO BRANCH VALUES (8,'Taste Factory','E-7 Islamabad','0518888888');  
INSERT INTO BRANCH VALUES (9,'Food Court','F-6 Islamabad','0519999999');  
INSERT INTO BRANCH VALUES (10,'Golden Grill','G-11 Islamabad','0511010101');
```

Output :

Results Explain Describe Saved SQL History

BRANCH_ID	BRANCH_NAME	ADDRESS	PHONE
1	Blue Moon	F-8 Islamabad	0511111111
2	Royal Taste	G-9 Islamabad	0512222222
3	Food Street	I-8 Islamabad	0513333333
4	Spice Hub	F-10 Islamabad	0514444444
5	Tandoori House	G-10 Islamabad	0515555555
6	Biryani Point	I-9 Islamabad	0516666666
7	Urban Eatery	H-8 Islamabad	0517777777
8	Taste Factory	E-7 Islamabad	0518888888
9	Food Court	F-6 Islamabad	0519999999
10	Golden Grill	G-11 Islamabad	0511010101



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2. EMPLOYEE

```
INSERT INTO EMPLOYEE VALUES (101,1,'Ali Khan','Chef',50000,DATE '2024-01-10');
INSERT INTO EMPLOYEE VALUES (102,1,'Ahmed Raza','Waiter',30000,DATE '2024-02-10');
INSERT INTO EMPLOYEE VALUES (103,2,'Usman Tariq','Chef',55000,DATE '2024-01-15');
INSERT INTO EMPLOYEE VALUES (104,2,'Bilal Ahmed','Waiter',32000,DATE '2024-03-05');
INSERT INTO EMPLOYEE VALUES (105,3,'Hassan Ali','Chef',60000,DATE '2024-01-25');
INSERT INTO EMPLOYEE VALUES (106,3,'Zeeshan Malik','Waiter',31000,DATE '2024-02-15');
INSERT INTO EMPLOYEE VALUES (107,4,'Saad Khan','Chef',58000,DATE '2024-03-01');
INSERT INTO EMPLOYEE VALUES (108,5,'Hamza Shah','Waiter',33000,DATE '2024-03-10');
INSERT INTO EMPLOYEE VALUES (109,6,'Owais Akram','Chef',62000,DATE '2024-04-01');
INSERT INTO EMPLOYEE VALUES (110,7,'Farhan Iqbal','Manager',70000,DATE '2024-04-05');
INSERT INTO EMPLOYEE VALUES (111,8,'Daniyal','Waiter',29000,DATE '2024-05-01');
INSERT INTO EMPLOYEE VALUES (112,9,'Kashif','Chef',64000,DATE '2024-05-10');
INSERT INTO EMPLOYEE VALUES (113,10,'Awais','Waiter',31000,DATE '2024-05-15');
INSERT INTO EMPLOYEE VALUES (114,1,'Imran','Cashier',28000,DATE '2024-06-01');
INSERT INTO EMPLOYEE VALUES (115,2,'Noman','Manager',75000,DATE '2024-06-05');
```

Output:

EMPLOYEE_ID	BRANCH_ID	EMPLOYEE_NAME	POSITION	SALARY	HIRE_DATE
101	1	Ali Khan	Chef	50000	10-JAN-24
102	1	Ahmed Raza	Waiter	30000	10-FEB-24
103	2	Usman Tariq	Chef	55000	15-JAN-24
106	3	Zeeshan Malik	Waiter	31000	15-FEB-24
108	5	Hamza Shah	Waiter	33000	10-MAR-24
112	9	Kashif	Chef	64000	10-MAY-24

3. CHEF

```
INSERT INTO CHEF VALUES (1,101,'Pakistani Food');
INSERT INTO CHEF VALUES (2,103,'Chinese Food');
INSERT INTO CHEF VALUES (3,105,'BBQ');
INSERT INTO CHEF VALUES (4,107,'Fast Food');
INSERT INTO CHEF VALUES (5,109,'Italian');
INSERT INTO CHEF VALUES (6,112,'Desi Food');
INSERT INTO CHEF VALUES (7,114,'Sea Food');
INSERT INTO CHEF VALUES (8,115,'Grill Special');
```

OUTPUT:

CHEF_ID	EMPLOYEE_ID	SPECIALTY
1	101	Pakistani Food
2	103	Chinese Food

4. WAITER

```
INSERT INTO WAITER VALUES (1,102,'Morning');
INSERT INTO WAITER VALUES (2,104,'Evening');
INSERT INTO WAITER VALUES (3,106,'Night');
INSERT INTO WAITER VALUES (4,108,'Morning');
INSERT INTO WAITER VALUES (5,111,'Evening');
INSERT INTO WAITER VALUES (6,113,'Night');
```

OUTPUT:

WAITER_ID	EMPLOYEE_ID	SHIFT
1	102	Morning
3	106	Night
4	108	Morning

5. SUPPLIER



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```
INSERT INTO SUPPLIER VALUES (1,1,'Fresh Foods Ltd','0300111111');
INSERT INTO SUPPLIER VALUES (2,2,'Meat Suppliers','0300222222');
INSERT INTO SUPPLIER VALUES (3,3,'Veggie Market','0300333333');
INSERT INTO SUPPLIER VALUES (4,4,'Spice Traders','0300444444');
INSERT INTO SUPPLIER VALUES (5,5,'Rice Distributors','0300555555');
INSERT INTO SUPPLIER VALUES (6,6,'Oil Suppliers','0300666666');
INSERT INTO SUPPLIER VALUES (7,7,'Bakery Goods','0300777777');
INSERT INTO SUPPLIER VALUES (8,8,'Beverage Co','0300888888');
INSERT INTO SUPPLIER VALUES (9,9,'Frozen Foods','0300999999');
INSERT INTO SUPPLIER VALUES (10,10,'Dairy Suppliers','0300101010');
```

OUTPUT:

SUPPLIER_ID	BRANCH_ID	SUPPLIER_NAME	PHONE
1	1	Fresh Foods Ltd	0300111111
2	2	Meat Suppliers	0300222222
3	3	Veggie Market	0300333333

3 rows returned in 0.02 seconds

[CSV Export](#)

6. CUSTOMER

```
INSERT INTO CUSTOMER VALUES (1,'Zain Ali','0311111111');
INSERT INTO CUSTOMER VALUES (2,'Ali Raza','0312222222');
INSERT INTO CUSTOMER VALUES (3,'Hamza','0313333333');
INSERT INTO CUSTOMER VALUES (4,'Usman','0314444444');
INSERT INTO CUSTOMER VALUES (5,'Bilal','0315555555');
INSERT INTO CUSTOMER VALUES (6,'Hassan','0316666666');
INSERT INTO CUSTOMER VALUES (7,'Saad','0317777777');
INSERT INTO CUSTOMER VALUES (8,'Ahmed','0318888888');
INSERT INTO CUSTOMER VALUES (9,'Daniyal','0319999999');
INSERT INTO CUSTOMER VALUES (10,'Farhan','0310101010');
```

OUTPUT:

CUSTOMER_ID	CUSTOMER_NAME	PHONE
1	Zain Ali	0311111111
2	Ali Raza	0312222222
3	Hamza	0313333333
4	Usman	0314444444
5	Bilal	0315555555
6	Hassan	0316666666

7. MENU_ITEM

```
INSERT INTO MENU_ITEM VALUES (1,'Burger','Fast Food',500);
INSERT INTO MENU_ITEM VALUES (2,'Pizza','Fast Food',1200);
INSERT INTO MENU_ITEM VALUES (3,'Biryani','Desi',300);
INSERT INTO MENU_ITEM VALUES (4,'Pasta','Italian',800);
INSERT INTO MENU_ITEM VALUES (5,'BBQ Platter','BBQ',1500);
INSERT INTO MENU_ITEM VALUES (6,'Fries','Fast Food',200);
INSERT INTO MENU_ITEM VALUES (7,'Shawarma','Fast Food',350);
INSERT INTO MENU_ITEM VALUES (8,'Steak','Continental',2000);
INSERT INTO MENU_ITEM VALUES (9,'Soup','Starter',250);
INSERT INTO MENU_ITEM VALUES (10,'Ice Cream','Dessert',150);
```

Output:



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ITEM_ID	ITEM_NAME	CATEGORY	PRICE
1	Burger	Fast Food	500
2	Pizza	Fast Food	1200
3	Biryani	Desi	300
4	Pasta	Italian	800
5	BBQ Platter	BBQ	1500
6	Fries	Fast Food	200

8. ORDERS

```
INSERT INTO ORDERS VALUES (1,1,1,DATE '2026-06-01',1800);
INSERT INTO ORDERS VALUES (2,2,2,DATE '2026-06-01',1200);
INSERT INTO ORDERS VALUES (3,3,3,DATE '2026-06-02',900);
INSERT INTO ORDERS VALUES (4,4,4,DATE '2026-06-02',1500);
INSERT INTO ORDERS VALUES (5,5,5,DATE '2026-06-03',2000);
INSERT INTO ORDERS VALUES (6,6,1,DATE '2026-06-03',700);
INSERT INTO ORDERS VALUES (7,7,2,DATE '2026-06-03',1100);
INSERT INTO ORDERS VALUES (8,8,3,DATE '2026-06-03',950);
INSERT INTO ORDERS VALUES (9,9,4,DATE '2026-06-03',600);
INSERT INTO ORDERS VALUES (10,10,5,DATE '2026-06-03',1300);
```

Output:

ORDER_ID	CUSTOMER_ID	WAITER_ID	ORDER_DATE	TOTAL_AMOUNT
1	1	1	01-JUN-26	1800
3	3	3	02-JUN-26	900
4	4	4	02-JUN-26	1500
6	6	1	03-JUN-26	700

9. ORDER_DETAILS

```
INSERT INTO ORDER_DETAILS VALUES (1,1,1,2);
INSERT INTO ORDER_DETAILS VALUES (2,1,3,1);
INSERT INTO ORDER_DETAILS VALUES (3,2,2,1);
INSERT INTO ORDER_DETAILS VALUES (4,3,3,3);
INSERT INTO ORDER_DETAILS VALUES (5,4,5,1);
INSERT INTO ORDER_DETAILS VALUES (6,5,8,1);
INSERT INTO ORDER_DETAILS VALUES (7,6,6,2);
INSERT INTO ORDER_DETAILS VALUES (8,7,4,1);
INSERT INTO ORDER_DETAILS VALUES (9,8,7,2);
INSERT INTO ORDER_DETAILS VALUES (10,9,9,1);
```

Output:

ORDER_DETAIL_ID	ORDER_ID	ITEM_ID	QUANTITY
2	1	3	1
1	1	1	2
4	3	3	3
5	4	5	1
7	6	6	2

10. PAYMENT

```
INSERT INTO PAYMENT VALUES (1,1,DATE '2026-06-01',1800,'Cash');
INSERT INTO PAYMENT VALUES (2,2,DATE '2026-06-01',1200,'Card');
INSERT INTO PAYMENT VALUES (3,3,DATE '2026-06-02',900,'Online');
INSERT INTO PAYMENT VALUES (4,4,DATE '2026-06-02',1500,'Cash');
INSERT INTO PAYMENT VALUES (5,5,DATE '2026-06-03',2000,'Card');
```




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```
INSERT INTO PAYMENT VALUES (6,6,DATE '2026-06-03',700,'Cash');
INSERT INTO PAYMENT VALUES (7,7,DATE '2026-06-03',1100,'Online');
INSERT INTO PAYMENT VALUES (8,8,DATE '2026-06-03',950,'Cash');
INSERT INTO PAYMENT VALUES (9,9,DATE '2026-06-03',600,'Card');
INSERT INTO PAYMENT VALUES (10,10,DATE '2026-06-03',1300,'Online');
```

Output:

PAYMENT_ID	ORDER_ID	PAYMENT_DATE	AMOUNT	PAYMENT_METHOD
3	3	02-JUN-26	900	Online
4	4	02-JUN-26	1500	Cash
6	6	03-JUN-26	700	Cash
1	1	01-JUN-26	1800	Cash

Part 3: Database Querying & Optimization (Minimum 10 Queries)

1. Basic SELECT Queries

```
SELECT * FROM BRANCH;
```

Output:

BRANCH_ID	BRANCH_NAME	ADDRESS	PHONE
1	Blue Moon	F-8 Islamabad	0511111111
2	Royal Taste	G-9 Islamabad	0512222222
3	Food Street	I-8 Islamabad	0513333333
4	Spice Hub	F-10 Islamabad	0514444444
5	Tandoori House	G-10 Islamabad	0515555555
6	Biryani Point	I-9 Islamabad	0516666666
7	Urban Eatery	H-8 Islamabad	0517777777
8	Taste Factory	E-7 Islamabad	0518888888
9	Food Court	F-6 Islamabad	0519999999
10	Golden Grill	G-11 Islamabad	0511010101

2. Show all employees

```
SELECT * FROM EMPLOYEE;
```

Output :

EMPLOYEE_ID	BRANCH_ID	EMPLOYEE_NAME	POSITION	SALARY	HIRE_DATE
101	1	Ali Khan	Chef	50000	10-JAN-24
102	1	Ahmed Raza	Waiter	30000	10-FEB-24
103	2	Usman Tariq	Chef	55000	15-JAN-24
106	3	Zeeshan Malik	Waiter	31000	15-FEB-24
108	5	Hamza Shah	Waiter	33000	10-MAR-24
112	9	Kashif	Chef	64000	10-MAY-24

3. Show all menu items

```
SELECT * FROM MENU_ITEM;
```

Output:

ITEM_ID	ITEM_NAME	CATEGORY	PRICE
1	Burger	Fast Food	500
2	Pizza	Fast Food	1200
3	Biryani	Desi	300
4	Pasta	Italian	800
5	BBQ Platter	BBQ	1500
6	Fries	Fast Food	200

2. JOIN QUERIES



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4. Employee with Branch name

```
SELECT E.Employee_Name, E.Position, B.Branch_Name  
FROM EMPLOYEE E  
JOIN BRANCH B  
ON E.Branch_ID = B.Branch_ID;
```

Output :

EMPLOYEE_NAME	POSITION	BRANCH_NAME
Ahmed Raza	Waiter	Blue Moon
Ali Khan	Chef	Blue Moon
Usman Tariq	Chef	Royal Taste
Zeeshan Malik	Waiter	Food Street
Hamza Shah	Waiter	Tandoori House
Kashif	Chef	Food Court

5. Orders with Customer and Waiter

```
SELECT O.Order_ID, C.Customer_Name, E.Employee_Name AS Waiter  
FROM ORDERS O  
JOIN CUSTOMER C ON O.Customer_ID = C.Customer_ID  
JOIN WAITER W ON O.Waiter_ID = W.Waiter_ID  
JOIN EMPLOYEE E ON W.Employee_ID = E.Employee_ID;
```

Output:

ORDER_ID	CUSTOMER_NAME	WAITER
1	Zain Ali	Ahmed Raza
3	Hamza	Zeeshan Malik
4	Usman	Hamza Shah
6	Hassan	Ahmed Raza

6. Order details with menu items

```
SELECT OD.Order_ID, M.Item_Name, OD.Quantity, M.Price  
FROM ORDER_DETAILS OD  
JOIN MENU_ITEM M ON OD.Item_ID = M.Item_ID;
```

Output:

ORDER_ID	ITEM_NAME	QUANTITY	PRICE
1	Biryani	1	300
1	Burger	2	500
3	Biryani	3	300
4	BBQ Platter	1	1500
6	Fries	2	200

3. AGGREGATE QUERIES

7. Total orders per customer

```
SELECT Customer_ID, COUNT(Order_ID) AS Total_Orders  
FROM ORDERS  
GROUP BY Customer_ID;
```

Output:

CUSTOMER_ID	TOTAL_ORDERS
1	1
6	1
4	1
3	1

4 rows returned in 0.01 seconds

[CSV Export](#)



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8. Total revenue of restaurant

```
SELECT SUM(Amount) AS Total_Revenue  
FROM PAYMENT;
```

Output:

Results Explain Describe Saved SQL History

TOTAL_REVENUE
4900

9. Average order amount

```
SELECT AVG(Total_Amount) AS Avg_Order_Value  
FROM ORDERS;
```

Output:

Results Explain Describe Saved SQL History

AVG_ORDER_VALUE
1225

1 rows returned in 0.00 seconds [CSV Export](#)

4. SUBQUERY

10. Customers who made highest order

```
SELECT Customer_Name  
FROM CUSTOMER  
WHERE Customer_ID IN (  
    SELECT Customer_ID  
    FROM ORDERS  
    WHERE Total_Amount = (  
        SELECT MAX(Total_Amount) FROM ORDERS  
    )  
);
```

Output:

CUSTOMER_NAME
Zain Ali

1 rows returned in 0.04 seconds [CSV Export](#)

View: Branch Sales Summary:

```
CREATE VIEW BRANCH_SALES AS  
SELECT  
    B.Branch_Name,  
    COUNT(O.Order_ID) AS Total_Orders,  
    SUM(P.Amount) AS Total_Revenue  
FROM BRANCH B  
JOIN EMPLOYEE E ON B.Branch_ID = E.Branch_ID  
JOIN WAITER W ON E.Employee_ID = W.Employee_ID  
JOIN ORDERS O ON W.Waiter_ID = O.Waiter_ID  
JOIN PAYMENT P ON O.Order_ID = P.Order_ID  
GROUP BY B.Branch_Name;
```

Output:



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View created.

0.04 seconds

TRIGGER

CREATE OR REPLACE TRIGGER UPDATE_ORDER_TOTAL
AFTER INSERT OR UPDATE OR DELETE ON ORDER_DETAILS
FOR EACH ROW

DECLARE

v_total NUMBER;

BEGIN

SELECT SUM(M.Price * OD.Quantity)

INTO v_total

FROM ORDER_DETAILS OD

JOIN MENU_ITEM M ON OD.Item_ID = M.Item_ID

WHERE OD.Order_ID = :NEW.Order_ID;

UPDATE ORDERS

SET Total_Amount = NVL(v_total, 0)

WHERE Order_ID = :NEW.Order_ID;

END;

/

Output:

Trigger created.

0.07 seconds

Part 4: Technical Report & Project Defense

Its already done as a complete report .